

10 Infectious diseases

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A pathogen causes transmissible disease; viruses replicate inside host cells, while bacteria may divide independently. Transmission can involve droplets, water or vectors, so prevention must target the route. Antibiotics act on bacterial structures/processes but do not treat viruses. Resistance evolves when a heritable variant survives exposure and reproduces.

Worked example: follow the reasoning

A vaccine reduces susceptible hosts; if enough transmission chains end, infection incidence in the community may fall even among some unvaccinated individuals.

Quick check · multiple choice

A Why do antibiotics not directly treat viral replication?

[1]

- A Viruses have more ribosomes
- B Viruses lack targeted bacterial processes
- C Antibiotics only work in air
- D All viruses are bacteria

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A bacterial population survives an antibiotic treatment. Explain selection of resistance in three linked steps.

[3]

Answers and explanation

Quick check: B. Viruses lack targeted bacterial processes

Question 1 [3 marks]

Heritable resistance variants pre-exist or arise by mutation [1]; antibiotic kills susceptible cells so resistant cells survive [1]; resistant cells reproduce and allele frequency rises [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

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